4-1

化实2101　　钱汝扬　　2021010850

解：设A、B的用户量分别为 a、b

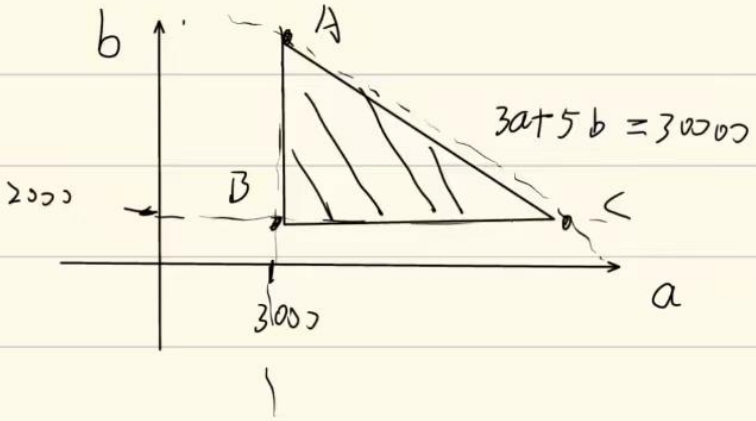
目标函数：$y = 2000a + 3000b$

约束条件：① $a \geq 3000$

② $b \geq 2000$

③ $3a + 5b \leq 30000$

目标函数可行域如下



可行域顶点：A：$(3000, 4200)$

B $(3000, 2000)$

C $C(\frac{20000}{3}, 2000)$

分别将A、B、C代入 y.

$y_A = 18600000$

$y_B = 12000000$

$y_C = 19333333$

选C 这对云云家，每周产A 6666.67L

产B 2000L

4-2

用图解法求解.

$$\text{Mins}=1000X_1+800X_2$$

$$\text{s.t.}\quad X1\geqslant 1$$

$$0.8X_1+X_2\geqslant 1.6$$

$$X_1\leqslant 2$$

$$X_2\leqslant 1.4$$

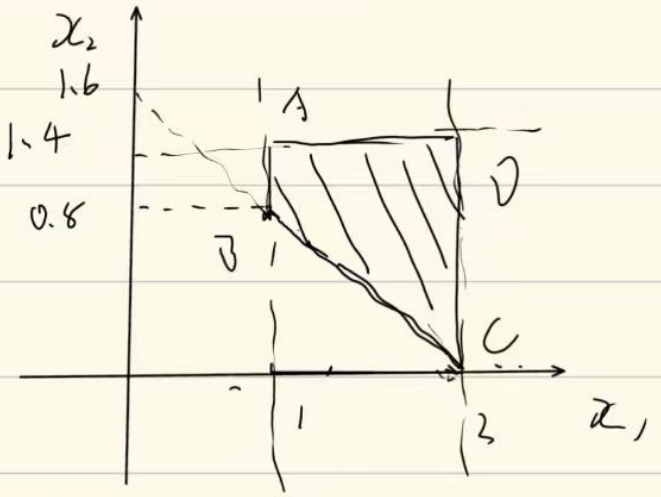
$$X_1\geqslant 0, X_2\geqslant 0.$$

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解. 可行域如下



A: (1, 1.4) B. (1. 0.8)

C. (2. 0) D. (2. 1.4)

$S_A = 2120$ $S_B = 1640$

$S_C = 2000$ $S_D = 3120$

min S = 1640

107%

4-3

非线性规划

1 $\min f(x) = (x_1 - 2)^2 + (x_2 - 1)^2$

s. t. $x_1 + x_2 + 5 = 0$

求 X1，X2

2 用牛顿法求无约束问题

$\min f(x) = x_1^2 + x_1x_2 + x_2^2 + 3x_1$，初始点 $x^0 = (1,1)^T$，迭代一次.

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4/ 解：令 $x_1' = x_1 - 2$ 、$x_2' = x_2 + 1$

则约束条件为 $x_1' + x_2' + 5 = 0$

建立带有罚函数的目标函数.

$F(x') = x_1'^2 + x_2'^2 + k(x_1' + x_2' + 5)^2 = 0$

对应一阶目标函数求极小值

$\frac{\partial F}{\partial x_1'} = 2x_1' + 2k(x_1' + x_2' + 5) = 0$

$\frac{\partial F}{\partial x_2'} = 2x_2' + 2k(x_1' + x_2' + 5) = 0$

化简得 $(x_1' + x_2') + k(x_1' + x_2' + 5) = 0$

当 $k \to \infty$，若要使上式成立.

有：$x_1' = -4$

$x_2' = -4$

解得 $x_1 = x_1' + 2 = -2$

$x_2 = x_2' + 1 = -3$

(2)

(2)

$$f(x) = f(x_1, x_2) = x_1^2 + x_1x_2 + x_2^2 + 3x_1$$

$$\frac{\partial f}{\partial x_1} = 2x_1 + x_2 + 3$$

$$\frac{\partial^2 f}{\partial x_1^2} = 2$$

$$\frac{\partial f}{\partial x_2} = x_1 + 2x_2$$

$$\frac{\partial^2 f}{\partial x_2^2} = 2$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} = 1$$

$$H(x) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

则 $H^{-1}(x) = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{bmatrix}$. $\nabla f(x) = \begin{bmatrix} 2x_1 + x_2 + 3 \\ x_1 + 2x_2 \end{bmatrix}$

$$\Delta x^0 = -H^{-1} \nabla f(x) = -\begin{bmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} 6 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \end{bmatrix}$$

$$x' = x^* + \Delta x^0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -3 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$f(x^*) = x_1^2 + x_1x_2 + x_2^2 + 3x_1$$

$$= 4 - 2 + 1 - 6$$

$$= -3$$

4-4（题目有错无解）

求解线性规划问题：

$$\begin{aligned} \max \quad & s = x_1 + 3x_2 + 4x_3 \\ s.t. \quad & 3x_1 + 5x_2 - 4x_3 \leqslant 10 \\ & -2x_1 + 3x_2 + x_3 \leqslant 5 \\ & x_j \geqslant 0, j = 1,2,3 \end{aligned}$$

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解：将原目标函数化为标准形，并在约束件上增加变量为剩余变量

$$\min -s = -x_1 - 3x_2 - 4x_3$$

$$s.t = 3x_1 + 5x_2 - 4x_3 + x_4 = 10 \quad (1)$$

$$-2x_1 + 3x_2 + x_3 + x_5 = 5 \quad (2)$$

$$x_j \geq 0 \quad j = 1,2,3$$

选择 x_4 x_5 为基变量，并设其余变量为 0

即：$x_4 = 10$ $x_5 = 5$

$$\bar{X} = (0,0,0,10,5)^T$$

$$B = (A_4, A_5) = I$$

$$C_B = (0,0)$$

$$y_{0j} = C_j - C_B B^{-1} \cdot A_j$$

$$y_{01} = C_1 - C_B B^{-1} \cdot A_1 = -1$$

$$y_{02} = -3 \qquad y_{03} = -4$$

$$y_{10} = C_B B^{-1} \cdot b = 1$$

			x_1	x_2	x_3	x_4	x_5
C_B	X_B	b	-1	-3	-4	0	0
0	x_4	3	3	5	-4	1	0
0	x_5	-2	-2	3	1	0	1

最优解满足：$y_j = C_j - C_B B^{-1} A_j \geq 0$

有 y_{0j} 不满足条件，初始可行解不是最优的

选择出现负检验数 y_{0j} 最小列为主列，

$q = 1$

最小比值 $\theta = \min\{y_{i0} / y_{i1} \mid y_{i0} > 0, 1 \leq i \leq n\}$

$P = 1$，修改单纯形表如下

迭代次数				x_1	x_2	x_3	x_4	x_5	θ	q
	C_B	X_B	b	0	$-\frac{5}{3}$	$-\frac{4}{3}$	$\frac{1}{3}$	6		
2	-1	x_1	$\frac{14}{3}$	1	$\frac{5}{3}$	$-\frac{4}{3}$	$\frac{1}{3}$	0	$\frac{14}{3}$	2
	0	x_5	$\frac{35}{3}$	0	$\frac{15}{3}$	$-\frac{5}{3}$	$\frac{2}{3}$	1		
	C_B	X_B	$\frac{117}{14}$	0	0	$-\frac{108}{14}$	$\frac{6}{14}$	$\frac{4}{14}$		
3	-1	x_1	$\frac{5}{14}$	1	0	$-\frac{17}{14}$	$\frac{3}{14}$	$-\frac{5}{14}$		3
	-3	x_2	$\frac{25}{14}$	0	1	$-\frac{5}{14}$	$\frac{2}{14}$	$\frac{3}{14}$		